

《Operating System》

Assignment-2

Major: SOFTWARE ENGINEERING

Name: MD TASNIMUL HASAN

Student ID: 202480090104

1. Definitions

Volatile Memory:
This is a type of temporary computer memory. It holds data only while the computer is powered on. As soon as you turn off the power or shut down the system, all data inside it is lost immediately. It is mainly used for fast, short-term tasks, such as RAM.

Non-Volatile Memory:
This is a permanent type of memory. It can retain data even when the power is turned off. You can shut down your computer completely, and the data will still be there when you turn it back on. Examples include hard drives, SSDs, and pen drives.

2. Why does volatile memory lose data?
Volatile memory (like RAM) stores data using electrical charges inside tiny capacitors. Each capacitor holds a charge to represent a 0 or 1. When power is removed, these capacitors cannot hold their charge anymore — the charge slowly drains away, and all data is lost. These memory cells also need constant refreshing, which is impossible without power.

Why does non-volatile memory NOT lose data?
Non-volatile memory (like flash memory / SSD) uses special transistors called **floating gate transistors**. These transistors can trap electrons inside an insulated layer even when power is off. Because the electrons stay trapped, the data remains saved. Later, when power is restored, the memory reads the trapped electron state to retrieve the data. Another type (like hard drives) uses magnetic storage, which also doesn't need power to stay intact.

Feature	Volatile Memory	Non-Volatile Memory
Data Retention	Loses data when power is removed	Retains data without power
Speed	Very fast (nanoseconds access time)	Slower (microseconds to milliseconds)
Cost per GB	Expensive (5$–10$ per GB for DRAM)	Cheaper (0.01$–0.10$ per GB for HDD/SSD)
Common Use Cases	RAM (system memory), CPU cache, registers	SSDs, HDDs, USB drives, memory cards, firmware storage
Power Dependency	Requires continuous power to hold data	No power needed for storage; some require power during writes/reads

Typical Devices	Desktop RAM, LPDDR in smartphones, GPU memory	NVMe SSDs, SD cards, BIOS flash, NAND/USB drives

Real-Life Examples: Volatile vs. Non-Volatile Memory

Volatile Memory

Example 1: RAM

- **Where it is used:** Laptops, desktops, smartphones, tablets
- **Its role in the functioning of the system:** RAM acts as the system's working memory. It temporarily stores the operating system, running applications, and active data so that the CPU can access them instantly. Without RAM, even the fastest processor would become extremely slow because it would have to read everything directly from the storage drive. When you open a program or file, it gets loaded from non-volatile storage into RAM for fast access.

Example 2: CPU Cache (L1, L2, L3)

- **Where it is used**: Inside the processor (CPU) of all computers, smartphones, and gaming consoles
- **Its role in the functioning of the system:** CPU cache functions as a high-speed holding area for the processor's most frequently used instructions and data. It predicts what the CPU will need next and stores a copy of it, preventing the CPU from waiting for the slower RAM. This dramatically speeds up processing and makes the system more responsive. The closer the cache is to the CPU cores, the faster it is, but also the smaller its capacity.

Non-Volatile Memory

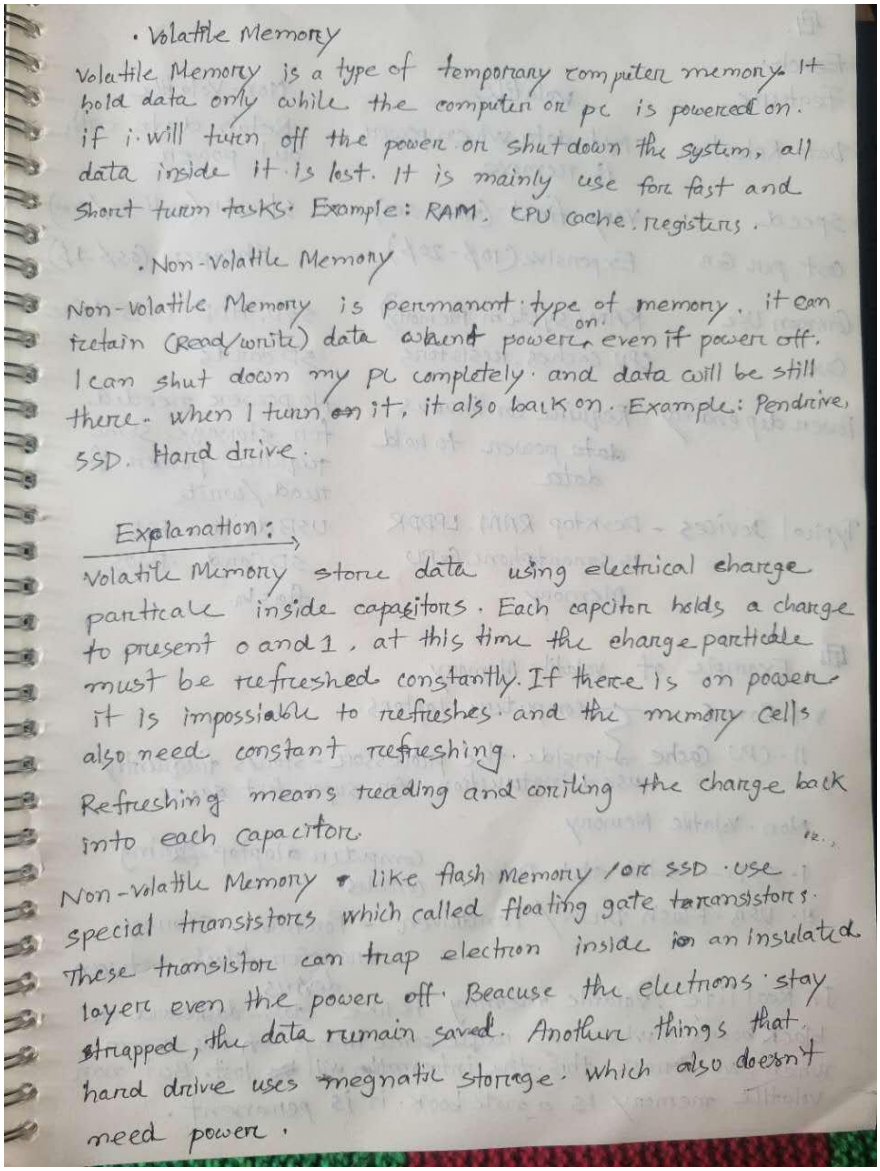
Example 1: SSD

- **Where it is used:** Laptops, desktops, gaming consoles (PS5, Xbox), servers
- **Its role in the functioning of the system**: SSD serves as the system's long-term storage. It permanently holds the operating system (Windows, macOS, Linux), installed

games, documents, photos, and videos. When you turn on the device, the operating system is copied from the SSD to the RAM to run. When you save a file, data moves from volatile RAM to the non-volatile SSD. Even after power is completely removed, all data on the SSD remains intact.

Example 2: USB Flash Drive

- **Where it is used:** Laptops, desktops, car stereos, printers, smart TVs, embedded devices
- **Its role in the functioning of the system:** The USB flash drive is used for portable data transfer and backup. It allows you to move files between completely different systems without an internet connection. It can also boot an operating system (like a Linux live USB), store media for a TV, update firmware on embedded devices, or keep important documents safe. Like SSD, it retains data permanently without needing power.



• Volatile Memory

Volatile Memory is a type of temporary computer memory. It hold data only while the computer or pc is powered on. if i will turn off the power on shut down the system, all data inside it is lost. It is mainly use for fast and short turm tasks. Example: RAM, CPU cache, registers.

• Non-Volatile Memory

Non-volatile Memory is permanent type of memory. it can retain (Read/write) data without power, even if power off. I can shut down my pc completely and data will be still there. when I turn on it, it also back on. Example: Pendrive, SSD. Hard drive.

Explanation:

Volatile Memory store data using electrical charge particle inside capacitors. Each capciton holds a charge to present 0 and 1, at this time the charge particle must be refreshed constantly. If there is on power it is impossiable to refreshes and the memory cells also need constant refreshing.

Refreshing means reading and coriling the charge back into each capacitor.

Non-volatile Memory like flash Memory /or SSD use special transistors which called floating gate transistors. These transistor can trap electron inside in an insulated layer even the power off. Beacuse the electrons stay trapped, the data remain saved. Another things that hard drive uses megnatic storage which also doesn't need power.

Feature	Volatile	Non-Volatile
Data Retention	lost data when power is remove	Retain data with out power
speed	Very first (nano-sec)	slower (mili-se/sec)
cost per GB.	Expensive (10-20)	Cheaper (0.5-1)
Common Use Cases	RAM (System Memory) CPU caches, resistors	SSD, HDD, USB drive SD cards
Power dependency	Require Continuous ~~data~~ power to hold data	No power needed for storage Some require power for read/write
Typical Devices	- Desktop RAM, LPDDR is smartphone GPU Memory	USB drive, SSD, SD Card, BIOS flash.

Example of volatile Memory

i. Ram { computers, laptops.

ii. CPU Cache } inside the processor - stores frequently used instructions for super-fast access.

Non-Volatile Memory

i. SSD — Solid state Drive — computer & laptop, gaming consoles.

ii. USB · Flash Drive/ Pendriven — Portable storage transfer files between devices.

In Real life, volatile memory is like class-dashboard or blackboard, when we write something on this. ~~we co~~ when we erase this. the information will ~~be~~ lost. But non volatile memory is a note book. it is pamanent.